High-Level Design (HLD) Structure

1.0 Introduction

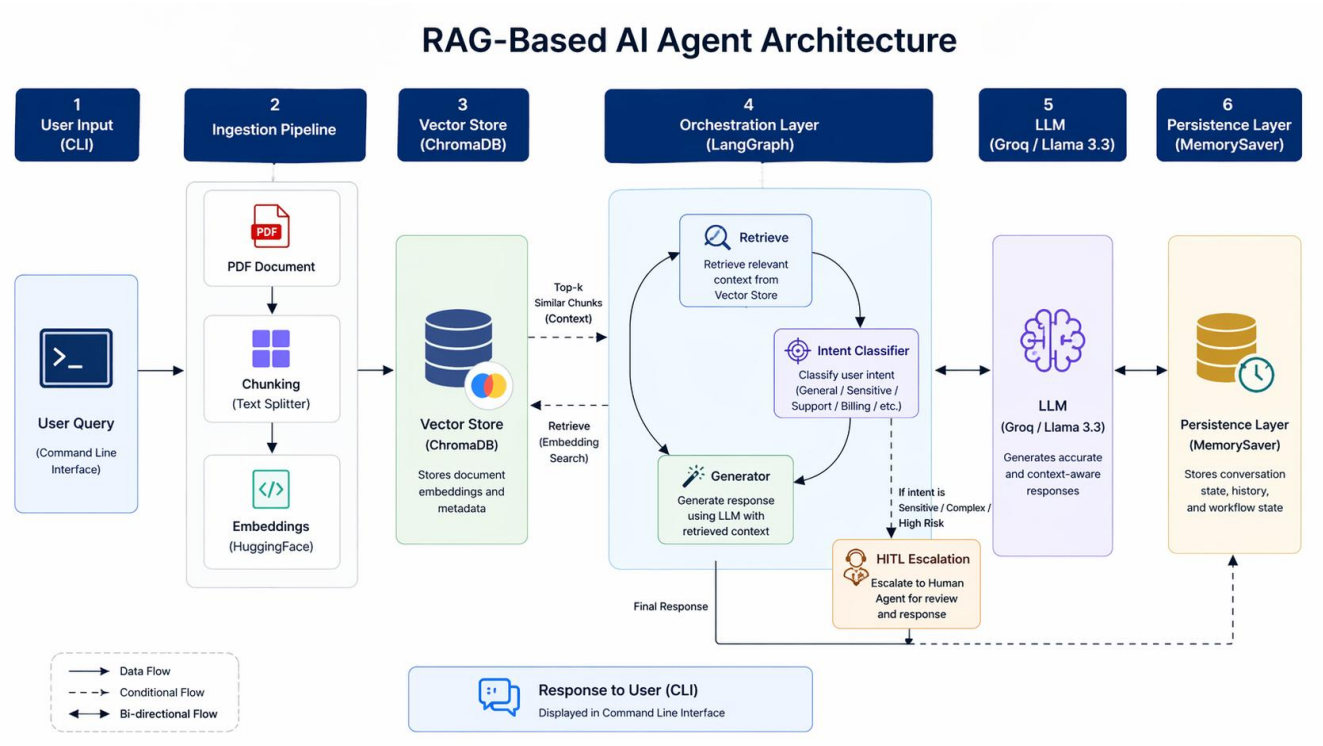
Project Name: AlphaTech Intelligent Support Agent

Objective: To develop a production-grade Retrieval-Augmented Generation (RAG) system that provides accurate technical support from a 27-page product manual while ensuring high-risk security queries are handled via Human-in-the-Loop (HITL) escalation.

2.0 System Architecture Overview

The system is built on a modular RAG architecture using **LangGraph** for workflow orchestration. Unlike linear chatbots, this system treats the support process as a state machine, allowing for conditional logic (routing) and session persistence (memory).

3.0 Architecture Diagram



4.0 Core Components

4.1 Document Ingestion Pipeline

- **Loader:** Uses PyPDFLoader to parse the 27-page AlphaTech PDF.
- **Splitter:** Implements RecursiveCharacterTextSplitter.
- **Strategy:** Chunks are set to **2000 characters** with a **200-character overlap**. This ensures that large technical specification tables are not fragmented, maintaining data integrity for the LLM.

4.2 Vector Storage & Embeddings

- **Embedding Model:** all-MiniLM-L6-v2 (HuggingFace). Selected for local inference to minimize latency and ensure data privacy.
- **Vector DB:** ChromaDB. Stores 384-dimensional vectors for semantic similarity searching.

4.3 Workflow Orchestration (LangGraph)

The system uses a **StateGraph** to manage the lifecycle of a query.

- **Stateful Memory:** Uses MemorySaver to track conversation history, allowing the agent to resolve pronouns (e.g., "What is its GPU?").
- **Nodes:** Independent functional units for Retrieval, Categorization, Generation, and Escalation.

5.0 Detailed Data Flow

1. **User Query:** User submits a technical or security-related question.
2. **Semantic Search:** The system retrieves the top 6 relevant context chunks from ChromaDB.
3. **Intent Classification:** The system analyzes the query for "Critical" keywords (e.g., Ransomware, Breach).
4. **Conditional Routing:**
    - **General Path:** If the intent is non-critical, the LLM generates a response based on the retrieved PDF context.
    - **Escalation Path:** If critical keywords are detected, the system bypasses generation and triggers a Human-in-the-Loop (HITL) notification.
5. **State Update:** The answer and question are saved to the history list within the graph's memory.

6.0 Technology Stack Choice

Component	Technology	Reason for Choice
Orchestration	LangGraphs	Supports complex cycles and persistence better than standard chains.
Vector DB	ChromaDB	Lightweight, local, and integrates natively with LangChain.
LLM	Llama 3.3 (Groq)	High-speed inference with 70B parameter reasoning for technical accuracy.
Embeddings	HuggingFace	Free, local, and highly efficient for CPU-based environments.

7.0 Scalability & Security

- **Security:** PII and sensitive security threats are automatically routed to human analysts, preventing the LLM from giving unauthorized instructions during a breach.
- **Scalability:** The architecture supports adding more PDF manuals without changing the underlying code by simply rerunning the ingestion script.